THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Ignoring air resistance in a model of a falling ball is acceptable for a stone. Reason (R): For a stone, the effect of air resistance is very small compared with its weight. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The mass of the Earth is about 6 x 10^24 kg and that of the Sun about 2 x 10^30 kg. The Sun is about ___ times heavier. 1

(a) 3 x 10^6 **(b)** 3 x 10^4 **(c)** 3 x 10^54 **(d)** 3 x 10^5

3. A scientist says 'The model fits in 99 of 100 trials, so the theory is proven'. A better statement is: 1

(a) the theory is false
(b) the theory is strongly supported but remains open to tests
(c) the theory is proven forever
(d) the data are wrong

4. Assertion (A): A scientific theory can never be changed. Reason (R): A theory is supported by a large amount of evidence. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A metre scale has a worn end and reads 0.3 cm at the start. A measured length reads 5.7 cm. The corrected length is: 1

(a) 5.4 cm **(b)** 0.3 cm **(c)** 6.0 cm **(d)** 5.7 cm

6. Which situation requires an exact value rather than an estimate? 1

(a) population of a country **(b)** number of stars in a galaxy
(c) height of a mountain in rough terms **(d)** dosage of a medicine for a child

7. Assertion (A): Estimation is useful in science. Reason (R): It gives a quick check on whether an answer is reasonable. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A stopwatch reads 20.4 s for 10 swings and each reading has an error of 0.2 s. The percentage error is about: 1

(a) 0.1 per cent **(b)** 1 per cent **(c)** 5 per cent **(d)** 10 per cent

9. An experimental result is 4.8 and the model gives 5.0. The first step in an honest investigation is to: 1

(a) change the 4.8 to 5.0
(b) drop the experiment
(c) ignore the model
(d) check the experiment and the assumptions in the model

10. Assertion (A): A measurement can never be exactly correct. Reason (R): Every instrument has a least count and every observation has some uncertainty. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The sides of a rectangle are measured as 12.0 cm and 8.0 cm with an uncertainty of 0.1 cm each. Find the area and the percentage uncertainty. 2

12. Why does a log-log or straight-line graph help in checking a mathematical relationship? 2

13. Convert 1 light year into kilometres, given speed 3 x 10^5 km/s and 1 year = 3.15 x 10^7 s. 2

14. A student's data for g from five trials are 9.7, 9.9, 9.8, 9.6, 9.8 m/s^2. Find the mean and the range. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student measures a pencil three times as 14.2 cm, 14.3 cm and 14.1 cm. Find the mean length and the greatest deviation from it. 3

16. Explain why a scientist should report measurement uncertainties along with results with an example. 3

17. Explain how a graph of time period against length for a pendulum is expected to look. 3

OR

The time for 30 oscillations of a pendulum is measured three times: 54.2 s, 54.6 s and 54.4 s. Find the mean time period and the percentage spread. 3

18. Estimate the mass of air in a classroom of size 8 m x 6 m x 3 m (density of air 1.2 kg per cubic metre). 3

OR

Explain how science and ethics are linked using the example of testing medicines. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an investigation to find the relation between the length and the time period of a simple pendulum: apparatus, variables, procedure, table, graph and conclusion. 5

OR

Explain how a chemist plans an experiment on the effect of temperature on the rate of dissolving of sugar, including fair-test conditions, data recording, graph and analysis of errors. 5

20. Discuss the strengths and limitations of the scientific method with five points and two examples of discoveries involving chance and persistence. 5

OR

The scientific community accepted heliocentrism after long debate. Describe how evidence, models and communication played a role. 5

END OF QUESTION PAPER — Total: 40 Marks